

4 Quantification

I calibrate a stationary reference economy representing 2007, before the subsequent decline in fertility. I use observations from the surrounding years to discipline fertility, housing, and household wealth. Housing moments use the national household sample in the 2005–2006 ACS; fertility moments come from the 2004 and 2006 CPS and 2003–2006 natality records, while aggregate wealth uses the 2005 and 2007 PSID. This initial steady state provides the starting point for the historical transition, in which changes in fertility preferences account for the subsequent path of births while the other structural parameters remain fixed.

I organize the model parameters into two groups. The first consists of parameters assigned from external evidence, measured directly in the data, or fixed as maintained restrictions. The second consists of parameters estimated internally by matching model moments to their empirical counterparts. I describe the two groups in turn, explain which moments are particularly informative about the internally estimated parameters, and then assess the model’s fit.

4.1 Parameters Set Outside the Estimation

I first describe the parameters assigned outside the estimation. Table 1 summarizes their values and restrictions.

Demographics and preferences. A model period is four years. Households enter adult life at age 18, retire at age 66, and live at most through age 85. Survival is certain before retirement and thereafter follows age-specific survival probabilities. Dependent children remain in the household for an expected 18 years, implying a maturity probability of $2/9$ per period. I set risk aversion to $\sigma = 2$ and retain the power equivalence scale with child weight $\omega = 0.7$ and exponent $\nu = 0.7$. The consumption weight is $\alpha_0 = 0.733$, measured using childless renters in the 2019–2023 CEX. These preference restrictions are held fixed in the estimation; they need not be measured in the base year.

Housing finance and transaction costs. I set the annual real return on liquid wealth to $i_{\text{ann}} = 0.02$, the proportional selling cost to $\psi^s = 0.06$, and the labor-income tax rate to $\tau_{\text{pay}} = 0.179$, retaining the external restrictions used in the housing-finance calibration (Greaney et al., 2025). Annual housing depreciation is $\delta_{\text{ann}} = 1.416\%$ and the annual property-tax rate is $\tau_{\text{ann}}^p = 1.060\%$. Mortgage debt cannot exceed a share $\phi = 0.80$ of the value of the house, so a purchase requires at least 20 percent equity. Purchase eligibility accounts for current earnings as well as existing resources. The model imposes no separate payment-to-income limit. Payroll-tax revenue finances retirement benefits through the stationary pension budget; the pension is therefore determined within each candidate equilibrium.

Income process. I impose a deterministic lifecycle profile of household earnings, e_a , normalized to average one over working ages. Within age, log earnings follow a persistent AR(1) process estimated from the PSID at the model’s four-year frequency. The process has persistence $\rho_z = 0.735$ and innovation standard deviation $\sigma_z = 0.484$ in log earnings. It has no fixed permanent household types and is approximated with 15 income states. Entrants draw from the stationary earnings-risk distribution.

Housing products and supply. Housing is indivisible and available in a limited assortment of types and sizes (Davis and Van Nieuwerburgh, 2015). I represent this lumpiness with five owner-occupied house sizes, $\{H_k\}_{k=1}^5 = \{2, 4, 6, 8, 10\}$ rooms. Renters choose housing continuously up to $\bar{h}^R = 6$ rooms. These menus summarize the tenure-specific distribution of rooms. I retain a housing-supply elasticity of $\eta = 0.63$; the supply intercept H_0 is estimated jointly with household parameters.

Entry wealth and bequests. Households enter at age 18 with heterogeneous nonhousing wealth and debt, constructed from the PSID wealth-to-family-income distribution of childless renters aged 18–24. The joint entry distribution retains the reference mapping between wealth and earnings. Bequest utility does not vary directly with the number of children. Children affect estate accumulation through their effects on housing needs, tenure, and saving. I fix the bequest wealth shift θ_1 at 1 percent of median annual gross earnings, or $\theta_1 = 0.008$ under the earnings normalization above.

Table 1: Parameters set outside the estimation

Parameter	Value	Evidence or restriction
Model period	4 years	Model timing
Entry, retirement, maximum ages	18, 66, 85	Lifecycle specification
Risk aversion σ	2	Maintained preference restriction
Equivalence scale (ω, ν)	(0.7, 0.7)	Maintained power scale
Consumption weight α_0	0.733	CEX 2019–2023, childless renters
Annual real return	2%	External return restriction
Annual depreciation; property tax	1.416%; 1.060%	Maintained housing-cost restrictions
Financed share ϕ	0.80	20% equity requirement
Sale cost ψ^s	6%	External transaction cost
Payroll tax	17.9%	External tax restriction
Income persistence ρ_z	0.735	Four-year PSID process
Income innovation SD σ_z	0.484	Four-year log-earnings innovation
Owner house sizes	2, 4, 6, 8, 10	Rooms
Largest rental unit	6	Rooms
Housing-supply elasticity η	0.63	Maintained supply restriction
Tenure taste-shock scale	0.005	Fixed in the working calibration
Entry wealth and debt	Heterogeneous	PSID childless renters aged 18–24
Bequest wealth shift θ_1	0.008	1% of median annual gross earnings

4.2 Internally Estimated Parameters

I estimate eight parameters jointly using moments describing childlessness, fertility timing and family size, housing consumption, tenure, and wealth accumulation. Separately, I choose the fertility utility scale ψ_0 so that mean children ever born among households aged 46 and above is 2.1 in the stationary economy. This is a normalization of the initial steady state, not an observed 2007 statistic or a period total fertility rate. Although the parameters are chosen jointly, I organize the discussion into economic blocks and describe which moments are most informative about each block.

Saving. The annual discount factor β_{annual} governs households’ willingness to carry resources across ages. Aggregate wealth relative to annual gross earnings is particularly informative about it.

The empirical target is a ratio of aggregate wealth to aggregate earnings, rather than the average of household wealth-to-earnings ratios.

Housing demand, tenure, and supply. I use housing-size differences by family composition to discipline housing demand. The room response around the first birth is particularly informative about the parenthood housing requirement h_P . The difference between larger and smaller families provides additional information about housing demand under the maintained equivalence scale. The floor is the same for all households with dependent children, rather than increasing with each additional child. Ownership rates and the ownership gap associated with recent parenthood are particularly informative about the owner housing-service multiplier χ .

Fertility and family size. Childlessness is particularly informative about the first-birth utility cost ξ . Mean age at first birth and the share of first births at age 30 or older help discipline first-birth taste dispersion κ_1 . The fraction of mothers with exactly one child helps discipline later-birth taste dispersion κ_C . These parameters are estimated jointly because the same family outcomes respond to the costs, timing, and utility value of children.

Bequests. The bequest parameters θ_0 and θ_1 govern the strength and wealth dependence of the bequest motive. With θ_1 fixed, annual bequests relative to aggregate wealth are particularly informative about θ_0 . This moment also responds to saving and tenure choices, so θ_0 is disciplined jointly with those blocks.

Table 2: Internally calibrated parameters and informative moments

Parameter	Interpretation	Particularly informative moments
β_{annual}	Annual discount factor	Wealth / gross earnings
ξ	First-birth utility cost	Childlessness
κ_1	First-birth taste dispersion	Mean first-birth age; share at 30+
κ_C	Later-birth taste dispersion	Exactly one child among mothers
h_P	Parenthood housing floor	First-birth room response; family room differences
χ	Owner housing-service multiplier	Ownership level; recent-parent gap
H_0	Housing-supply scale	—
θ_0	Bequest-motive strength	Annual bequests / wealth
ψ_0	Utility from children	Mean children ever born: 2.1 normalization

4.3 Target Moments and Fit

The target system combines children-ever-born distributions from the CPS, first-birth timing from natality records, tenure and housing-size moments from the ACS, housing responses around childbirth from the PSID event study, and aggregate wealth from the PSID. Table 3 reports the targets for the 2007 reference economy.

Table 3: Calibration targets for the 2007 steady state

Moment	Target	Model
Mean children ever born, ages 46+ (normalization)	2.10	—
Childlessness, women 40–44 (%)	19.83	—
Exactly one child among mothers, 40–44 (%)	21.37	—
Mean age at first birth (years)	25.98	—
First births at age 30+ (%)	24.93	—
Mean occupied rooms, capped at nine	5.608	—
Ownership, household heads 30–55 (%)	67.63	—
First-birth room response, $-3/-2$ to $+3/+4$ (rooms)	0.600	—
Rooms: 3+ versus 1–2 resident children	0.385	—
Recent-parent ownership gap (percentage points)	12.76	—
Wealth / annual gross earnings	6.927	—
Annual bequests / wealth (%)	0.88	—

References

- Morris A. Davis and Stijn Van Nieuwerburgh. Housing, Finance, and the Macroeconomy. In *Handbook of Regional and Urban Economics*, volume 5, pages 753–811. Elsevier, 2015. ISBN 978-0-444-59533-1. doi: 10.1016/B978-0-444-59531-7.00012-0.
- Brian Greaney, Andrii Parkhomenko, and Stijn Van Nieuwerburgh. Dynamic Urban Economics. *NBER Working Paper*, 2025.